Kyungtae Lee

Contact Information

Ph.D. Economics
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Education

Ph.D. Economics, City University of New York, Graduate Center	Present
M.S. Economics, University of Wisconsin, Madison	2019
B.A. Economics, The Pennsylvania State University, University Park	2016
B.A. Mathematics, The Pennsylvania State University, University Park	2016

Research interests

Energy Economics, Environmental Economics, Applied Econometrics, Development Economics

Teaching Field

Intro Statistics, Principles of Microeconomics, Intermediate Macroeconomics, Financial Economics, Quantitative Methods for Economics

Job Market Paper

Clean Energy Access and Home Production Improvement: Labor Supply and Gender Difference (**Top 4 Finalist for the Best Student Paper Award for 2025 USAEE Conference**)

The study examines the impact of adopting a clean energy cookstove on the intensive and extensive margin of labor supply, addressing the endogeneity issue. Using nationally representative data from Nepal, I found that adopting clean stoves increases men's labor participation and workdays, while reducing women's. Men worked more hours with clean stoves, whereas women's hours were negative but did not change significantly after adopting clean cookstoves.

Published

External Validity in an Instrumental Variable Setting
(with *Alexander Kwon*) Evaluation Review (2025)

We studied the external validity of instrumental variable estimation. The key assumption we impose for external validity is conditional external unconfoundedness among compliers, which means that the treatment effect and target selection are independent among compliers conditional on covariates. Using a case study on solid-fuel use and women's cooking time across six countries, we found no violations of this assumption in Ethiopia, Honduras, Kenya, and Zambia; identified limited external validity in Cambodia and Nepal.

Geographical Factors Influencing Household Cookstove Choices in Nepal
(with *Alexander Kwon*) Social Sciences & Humanities Open (2025)

We employ a logit model to analyze the factors, especially geographical characteristics, influencing households' choices regarding LPG stove adoption and generate slope and elevation data. We find that the estimates of the average slope are robust and statistically significant, but those for the average elevation are sensitive to model specifications and smaller compared to the slope. We find that the Kathmandu region is important in the analysis, and slope and elevation have nonlinear effects. Additionally, we show *that geographical factors are similarly important across different household expenditure quintile groups, except for the lowest*

Working Paper

Carbon Pricing and Heterogeneous Abatement Responses: Evidence from Seven U.S. Cities (with *Maggie Rong Hu* and *Jun Yoo*)

This paper examines whether carbon-pricing policies drive deeper emissions cuts than disclosure-only programs. Using building-level data from seven U.S. cities, we compare New York City's Local Law 97, which is the penalty based policy, with disclosure-focused approaches elsewhere. A difference-in-differences analysis shows modest but significant average reductions, with wide variation across buildings. We find an inverted U-shaped response to expected penalties. Overall, results suggest current penalties fall short of the response-maximizing level.

The Role of Accessibility in the Adoption of Clean Energy Stoves

I examine how accessibility influences LPG stove demand and provide empirical evidence that household transportation ownership positively affects the adoption of LPG stoves. Using data from Nepal's Multi-Tier Framework Survey (MTF), I apply Propensity Score Matching to address self-selection bias; the Average Treatment Effect on the Treated (ATET) is positive and significant across model specifications. To control community-level information, I merge MTF data with the Household Risk and Vulnerability Survey (HRVS) at the district level. Although this reduces the sample size, the results remain robust, with a larger and still significant ATET.

Work In Progress

From One LATE to Another: Machine Learning and the External Validity of IV Estimates (with *Alexander Kwon*)

Conference Presentation

Clean Energy Access and Home Production Improvement: Labor Supply and Gender Difference

- IAEE session at 2026 ASSA Conference
- 2025 USAEE/IAEE North American Conference

External Validity in an Instrumental Variable Setting in Practice: Machine Learning Approach

- 2026 Africa Meeting of the Econometric Society (Accepted)
- 2026 ASSA Conference (Poster)

External Validity in an Instrumental Variable Setting

- 2025 ASSA Conference (Lightning Session)

Teaching Experience

Brooklyn College

Adjunct Instructor, Financial Economics — Spring 2026

Adjunct Instructor, Intro Business Statistics — Spring 2026

Baruch College

Adjunct Instructor, Introduction to Microeconomics — Spring 2025–Spring 2026

Average evaluation: 4.00/5.00

Class size: 75–85 students

The City College of New York

Teaching Assistant, Principles of Microeconomics — Fall 2025

Average evaluation: 4.10/5.00

Adjunct Instructor, Principles of Statistics — Spring 2025

Average evaluation: 5.00/5.00

Adjunct Instructor, Intermediate Macroeconomics — Spring 2022 – Spring 2023; Summer 2025

Average evaluation: 3.85/5.00

The Graduate Center

Teaching Assistant, Math for Economist (Econ Ph.D. program) — Fall 2022

Hunter College

Instructor, Math Camp (Econ Master's program) — Summer 2022

New York University

Teaching Assistant, Mathematics for Economists (Master's Program) — Fall 2021

Average evaluation: 4.21/5.00

Work Experience

Short-term Consultant, The World Bank, Washington DC — Sept 2023 - Dec 2024

- Analyzed electricity demand, affordability, and access patterns using household data.
- Supported cross-country comparisons of energy demand and access conditions to inform energy policy and market assessments.
- Cleaned, harmonized, and organized large multi-country datasets for analysis and reporting.
- Assisted in evaluating how changes in prices, access, and technology adoption affect household energy use.
- Designed and refined household survey instruments to support data collection on energy demand and affordability.
- Contributed data analysis and figures for internal briefs and policy reports used by energy sector teams.

Scholarship and award

Dennis J. O'Brien USAEE Best Student Paper Finalist Award — 2025
Tuition Fellowship, CUNY Graduate Center — 2019 - 2024
Sallie Mae Awards, CUNY Graduate Center — 2021

Technical Skills

STATA, R, Python, Julia, MATLAB, Git, SQL, Latex

Languages

English(Fluent), Korean(Native)

References

Wim P.M. Vijverberg (Advisor)
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